

LINUX

Putty

Mobaxterm - IT is used for remote computing to manage remote systems.

sshubuntu@[IP_address]- here its is used to securely connect to remote systems over a network. Ubuntu@ specifies the username you want to log in ip address always be a public ip address

This command will open the ssh connection to remote sever of the ip

If there is a primary key then you have to type

ssh -i my-key.pem ubuntu@[ip]. HERE my-key means the location of the key and the key

Bash- Bash is a command shell where we can to command line scripting using scripting language in Operation system like linux.

Commands

Command	Description
uname -a	Show kernel version and system architecture
hostname	Show system hostname
uptime	Show system uptime and load average
whoami	Show current logged-in user
id	Show user ID and group info
top	Real-time process and resource usage
htop	Interactive process viewer (if installed)

Command	Description
vmstat 1	Show CPU, memory, I/O stats every second
iostat -xz 1	Show detailed disk I/O stats (requires sysstat)
free -h	Show memory usage in human-readable format
df -h	Show disk usage in human-readable format
du -sh *	Show size of each file/folder in current directory
lsblk	List block devices
blkid	Show block device UUIDs and types
mount	Show mounted filesystems
who	Show who is logged in
w	Show active logged-in users and processes
users	Show all logged-in usernames
groups	Show groups of current user
sudo adduser	Add new user
sudo passwd	Set password for user
sudo userdel -r	Delete user and home directory
sudo groupadd	Create new group
sudo usermod -aG	Add user to group
id	Show UID, GID, and groups of a user
pwd	Print current directory
ls -lah	List files with details and hidden files
cd /path/to/dir	Change directory
mkdir dir	Create a new directory
rm -rf dir	Delete directory and contents

Command	Description
touch file	Create empty file
cp src dest	Copy file or directory
mv src dest	Move or rename file/directory
find /path -name "file"	Search for file by name
grep "pattern" file	Search for text in file
grep -R "pattern" /path	Search recursively in directory
chmod 755 file	Change file permissions
chown user:group file	Change file owner and group
stat file	Show detailed file info
umask	Show default permission mask
ps aux	List all processes
ps -ef grep	Find process by name
kill	Kill process by PID
kill -9	Force kill process
pkill	Kill process by name
systemctl status	Check service status
systemctl restart	Restart service
systemctl enable	Enable service on boot
journalctl -xe	View detailed system logs
ip a	Show network interfaces and IP addresses
ifconfig	Show network config (deprecated)
ping -c 4 host	Ping host 4 times
curl -I url	Fetch HTTP headers
wget url	Download file from URL
netstat -tulnp	Show listening ports (deprecated)

Command	Description
ss -tulnp	Show listening ports (modern)
traceroute host	Trace route to host
dig domain	DNS lookup for domain
nslookup domain	DNS lookup for domain
telnet host port	Test TCP connection
nc -zv host port	Check if port is open
scp file user@host:/path	Copy file to remote server
rsync -avz src dest	Sync files between local/remote
sudo apt update && sudo apt upgrade	Update package list and upgrade packages
sudo apt install	Install package
sudo apt remove	Remove package
sudo yum install	Install package
sudo yum remove	Remove package
sudo yum update	Update all packages
tail -f /var/log/syslog	View live system log
tail -f /var/log/messages	View live messages log
tail -n 100 file.log	Show last 100 lines of file
less file.log	View file with scroll
grep "error" /var/log/syslog	Search 'error' in system log
dmesg less	View kernel log
df -h	Show disk usage
du -sh /path/*	Show size of all files/folders in path
lsblk	List storage devices

Command	Description
<code>mount /dev/sdX /mnt</code>	Mount device to /mnt
<code>umount /mnt</code>	Unmount /mnt
<code>fdisk -l</code>	List disk partitions
<code>mkfs.ext4 /dev/sdX</code>	Format disk with ext4 filesystem
<code>ssh user@host</code>	Connect to remote server
<code>ssh -i key.pem user@host</code>	Connect with private key
<code>ssh-copy-id user@host</code>	Copy public key to server
<code>tar -czvf archive.tar.gz file</code>	Create compressed tar.gz
<code>tar -xzf archive.tar.gz</code>	Extract tar.gz
<code>zip file.zip file</code>	Create zip archive
<code>unzip file.zip</code>	Extract zip archive
<code>nano file</code>	Edit file in nano
<code>vi file</code>	Edit file in vi
<code>vim file</code>	Edit file in vim
<code>crontab -l</code>	List cron jobs
<code>crontab -e</code>	Edit cron jobs
<code>systemctl status cron</code>	Check cron service status
<code>docker ps</code>	List running containers
<code>docker images</code>	List images
<code>docker exec -it /bin/bash</code>	Enter container shell
<code>docker logs</code>	View container logs
<code>kubectl get pods</code>	List pods

Command	Description
kubectl describe pod	Show pod details
kubectl logs	View pod logs
kubectl exec -it -- bash	Enter pod shell
git clone	Clone repository
git status	Show repo status
git add .	Stage all changes
git commit -m "msg"	Commit changes
git push	Push to remote
git pull	Pull latest changes

CREDENTIALS COMMAND

adduser [username] — Ubuntu

useradd -m [username] — CentOS

gpasswd -a [username] [group] → add a user to a group so that user gets same permission which a group has.

chown :[groupname] [path of directory] /home/app

chmod 770 [directory]

cd /etc/ssh/sshd_config → adding name in AllowUsers line will give user permission for SSH

User Management

passwd [username] → Set/change a user's password.

id [username] → Show user's UID, GID, and group memberships.

groups [username] → List all groups a user belongs to.

deluser [username] (Ubuntu) / userdel -r [username] (CentOS) → Remove a user (with home directory).

`usermod -aG [group] [username]` → Add user to a secondary group.

`usermod -s /bin/bash [username]` → Change user's login shell.

`who / w` → Show who is logged in.

`last` → Show login history.

Group Management

`groupadd [groupname]` → Create a new group.

`groupdel [groupname]` → Delete a group.

`getent group [groupname]` → Show group info.